



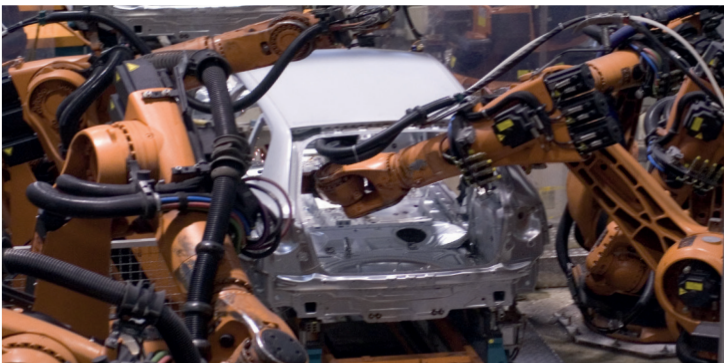
A-Trode®

The universal cap electrode for
resistance welding of light-to-medium
gauge coated and uncoated steels

A-Trode® production

Produced by Luvata's proprietary billet casting process, A-Trode's freedom from oxygen allows us to alloy the copper with the optimum levels of chrome and zirconium. Electrodes are cold headed or machined to suit the required shape.

Luvata has a long history of supplying high quality CuCrZr (A-Trode) electrodes.

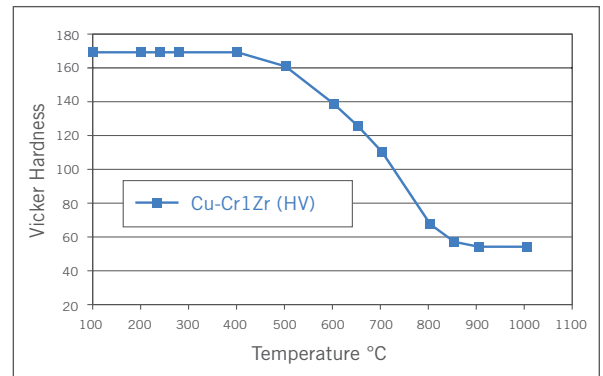


About Luvata

Luvata is a world leader in copper manufacturing and related engineering services. It brings together people, innovation and technology to create solutions in areas such as e-mobility, healthcare, and in electrification & the green transition. The company's success is driven by innovation, lean production and value-added sales, and its vision is to provide copper solutions which benefit people, society and the earth. Employing over 1,400 staff on four continents, Luvata works in partnership with customers such as ABB, CERN and Siemens. Luvata is a group company of Mitsubishi Materials Corporation.

Quality Properties

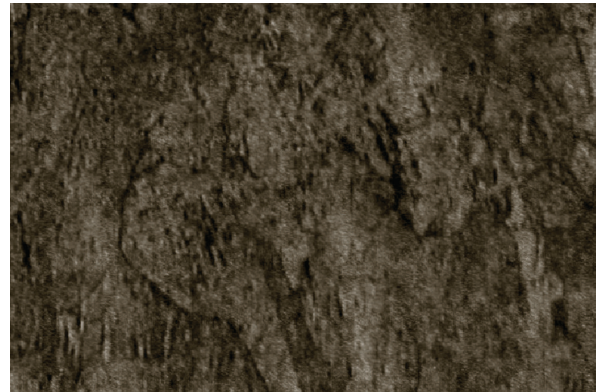
Alloy	C18150 CuCrZr, EN ISO 5182 A2/2, DIN 17666 Wn 2.1293, RWMA Class II
Physical material properties at 20°C	Mass Specific heat Thermal conductivity Expansion coefficient (20-300°C) Softening temperature 8,9g/cm³ 376 J/kg.K 320 W/m.K 17,0 x 10⁻⁶ m/mK min. 500°C
Dimensions and tolerances	To ISO 5821 or other standards as required. Special electrodes to customer drawing.
Packaging	Most items in cartons of 500 pieces.
Documentation	Acceptance test certificate EN 10204 3.1 possible if desired against a charge.
Area of application	Male and female resistance welding electrodes Backing dies - series and indirect Seam welding wheels Projection welding electrodes



A-Trode® metallurgy



Conventional CuCrZr grain structure



Luvata's CuCrZr fine grain structure

Mechanical Properties

Form of supply	Tensile strength [N/mm ²]	0.2% offset yield strength [N/mm ²]	Elongation AS [%]
Electrodes	≥ 490	≥ 430	≥ 15

Physical Properties

Hardness at ambient temperature	Minimum 75 HRB
Conductivity	Minimum 80% IACS

Errors and omissions excepted. Values given are published industry material standards.

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Cupal®

A cost effective, high-performance cap electrode

The Cupal cap is a cold-formed alloy of dispersion strengthened copper (DSC) with aluminum oxide. It consistently outperforms copper chrome and copper chrome zirconium electrodes in resistance to annealing, consistent electrical conductivity, electrode life and lower maintenance costs.

Cupal® resistance-welding electrodes

by **LUVATA**



Advantages of Cupal welding electrodes:

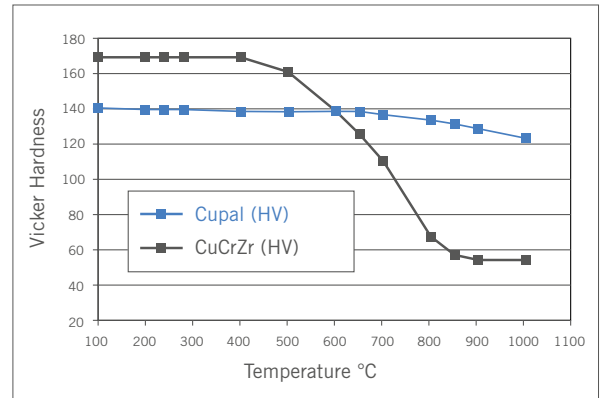
- longer weld life
- anti-stick characteristics
- reduces energy requirements
- works on all steels
- resists mushrooming for improved up-time

About Luvata

Luvata is a world leader in metal solutions manufacturing and related engineering services to industries such as renewable energy, automotive, healthcare, and power generation and distribution. The company's continued success is attributed to its longevity, technological excellence and strategy of building partnerships beyond metals. Employing approximately 1,400 staff in 6 countries, Luvata works in partnership with customers such as ABB, CERN, Siemens and Toyota. Luvata is a group company of Mitsubishi Materials Corporation.

Quality Properties*

Alloy	C15725 CuAl ₂ O ₃ , EN ISO 5182 C20/1, RWMA Class 20
Chemical composition	Aluminum 0.25% by weight as Al ₂ O ₃ Copper balance
Physical material properties at 20°C	Density 8.86 g/cm ³ Thermal conductivity 344 W/m.K Expansion coefficient (20-150°C) 16.6 x 10 ⁻⁶ m/mK Electrical conductivity 87% IACS Softening temperature 1083°C
Dimensions and tolerances	To ISO 5821 or other standards as required. Special electrodes to customer drawing.
Packaging	Most items in cartons of 500 pieces.
Documentation	Acceptance test certificate EN 10204 3.1 B possible if desired against a charge.
Area of application	Male and female resistance welding electrodes



Hardness at high temperature



Grain structure of DSC with aluminum oxide

Mechanical Properties*

Form of supply	Tensile strength MPa	Yield strength MPa	Elongation [%]	Hardness HV
Electrodes	441 - 489	365 - 413	18 - 23%	140

Physical Properties*

Hardness at ambient temperature	76 HRB
Electrical conductivity	87% IACS

*Errors and omissions excepted. Values given are industry standards. Actual properties will vary dependent upon amount of cold work.

Traceability

All materials are fully traceable. A Cupal electrode can be recognized by its four (4) pair of knurls evenly spaced around the periphery of the electrode.



Luvata Cupal electrodes

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Nitrode®

No other cap electrode can match Nitrode® for its cost-effective, high quality performance in resistance-welding applications

The Nitrode cap is a cold-formed alloy of copper dispersion strengthened with aluminum oxide. It consistently outperforms copper chrome and copper chrome zirconium electrodes in resistance to annealing, consistent electrical conductivity, electrode life and lower maintenance costs.

Longer weld life

Nitrode caps last longer than conventional electrodes by resisting annealing. The contact surface of a resistance welding electrode reaches temperatures of up to 900°C during the welding process. Nitrode electrodes resist the mushrooming effect on the contact surface of the electrode, allowing more welds before tip dressing, heat stepping or tip change.

Non-stick

Nitrode caps reduce sticking of electrodes on galvanized steel and other coated metals. The refractory qualities of Al_2O_3 reduce the infusion of liquid and gaseous zinc into the copper matrix.

Reduces energy requirements

Nitrode requires lower current when used on both sides of the weldment. Current settings on your welder can be reduced by up to 10% from conventional settings, with no loss of weld quality. Nitrode allows for smoother startups with no warm ups, meaning fewer electrode changes and fewer interruptions for less maintenance downtime than other conventional electrodes.

Works on all steels

Nitrode has demonstrated superior welding performance for a variety of steels, including HSS, HSLA, micro-alloyed, nitrogenized, low-carbon, electrolytically zinc-coated, galvanized and others.

Nitrode® cuts your company's costs

Nitrode resists mushrooming for improved up-time from reduced tip changes, less tip-dressing, and lower current settings will increase your company's productivity, and cut costs.

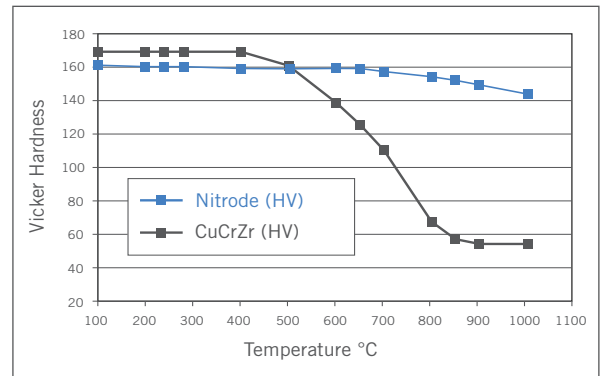


About Luvata

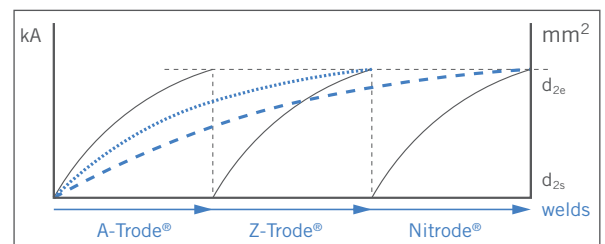
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Quality Properties*

Alloy	C15760 CuAl ₂ O ₃ , EN ISO 5182 C20/1, RWMA Class 20
Chemical composition	Aluminum 0.60% by weight as Al ₂ O ₃ Copper balance
Physical material properties at 20°C	Density 8.81g/cm ³ Thermal conductivity 322 W/m.K Expansion coefficient (20-150°C) 16.6 x 10 ⁻⁶ m/mK Electrical conductivity 78% IACS Softening temperature 1083°C
Dimensions and tolerances	To ISO 5821 or other standards as required. Special electrodes to customer drawing.
Packaging	Most items in cartons of 500 pieces.
Documentation	Acceptance test certificate EN 10204 3.1 B possible if desired against a charge.
Area of application	Male and female resistance welding electrodes Backing dies - series and indirect Projection welding electrodes



Hardness at high temperature



Heat stepping

Mechanical Properties*

Form of supply	Tensile strength MPa	Yield strength MPa	Elongation [%]	Hardness HV
Electrodes	489 - 537	393 - 441	18 - 23%	160

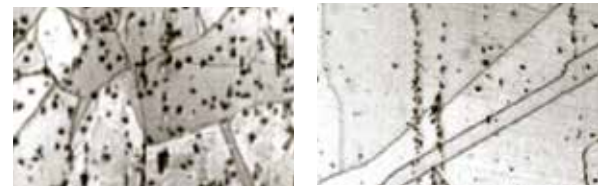
Physical Properties*

Hardness at ambient temperature	83 HRB
Electrical conductivity	78% IACS

*Errors and omissions excepted. Values given are industry standards. Actual properties will vary dependent upon amount of cold work.

Traceability

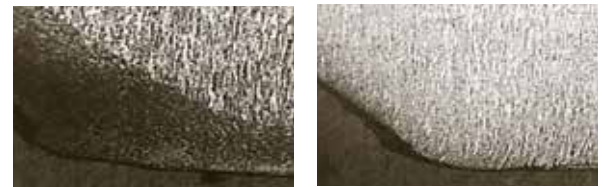
All materials are fully traceable. Nitrode electrodes can be recognized by their single knurls around the periphery of the electrode.



CuCrZr before and after annealing at 900°



Nitrode before and after annealing at 900°



CuCrZr after 1200 welds

Nitrode after 1200 welds

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Z-Trode®

The ideal cap electrode for resistance welding of light-to-medium gauge coated and uncoated steels and aluminum alloys

The Z-Trode® cap electrode is cold-formed from CDA Alloy 15000 zirconium copper. The combination of advanced copper metallurgy and our cold-forming technology gives you an electrode that outperforms conventional electrodes.

Prevents sticking

Chrome in a conventional electrode does not prevent zinc from alloying with copper to create a brass intermetallic across the weld face, this intermetallic diminishes the weld nugget size and degrades the weld quality. Z-Trode is an alloy of pure zirconium and oxygen-free copper. The zirconium discourages the alloying effect on the electrode weld face and prevents sticking. This increases productivity without special dressings or increased power requirements to weld coated materials.

Resists mushrooming

When compared to CuCrZr and CuCr alloys, the zirconium-copper alloy resists mushrooming and wear due to its superior conductivity. Z-Trode's conductivity allows it to be used with lower current settings and less heat, greatly improving the alloy's resistance to mushrooming.

Requires less energy

In comparison to CuCrZr and CuCr alloys, Z-Trode requires lower current due to its higher electrical conductivity, when used on both sides of the weldment. This results in longer electrode life, more consistent welds and energy savings. Weld current setting can be reduced up to 20% from the conventional CuCrZr and CuCr settings, with no loss in weld integrity.

Reduces downtime

Overall maintenance of Z-Trode caps is much less than with conventional electrodes, increasing welding process and production efficiencies. Z-Trode caps require no warm ups or preparation following electrode changes, which in turn makes the Z-Trode cap the most cost effective with its extended life expectancy and lower energy requirements.



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Quality Properties*

Alloy	C15000 CuZr, EN ISO 5182 A2/4, DIN 17666 Wn 2.1580, RWMA Class I
Chemical composition	Zirconium 0.15% Cu balance
Physical material properties at 20°C	Mass 8.89g/cm³ Specific heat 385 J/kg.K Thermal conductivity 367 W/m.K Expansion coefficient (20-300°C) 16.9 x 10⁻⁶ m/mK Electric conductivity min. 53 S/m Solution annealed and hardened min. 92% IACS Softening temperature 1083°C
Dimensions and tolerances	To ISO 5821 or other standards as required. Special electrodes to customer drawing.
Packaging	Most items in cartons of 500 pieces.
Documentation	Acceptance test certificate EN 10204 3.1 B possible if desired against a charge.
Area of application	Male and female resistance welding electrodes Backing dies - series and indirect Projection welding electrodes

Mechanical Properties*

Form of supply	Tensile strength [N/mm ²]	0.2% offset yield strength [N/mm ²]	Elongation AS [%]	Hardness HV
Electrodes	≥ 420	≥ 379	≥ 12	≥ 132

Physical Properties*

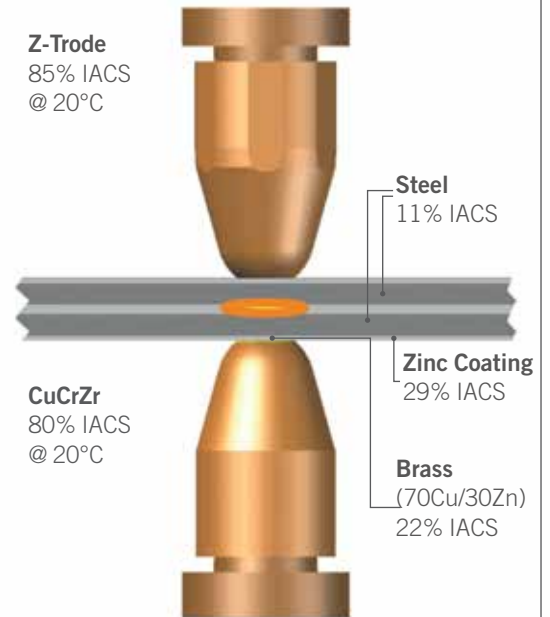
Hardness at ambient temperature	Minimum 65 HRB
Conductivity	Minimum 85% IACS

*Errors and omissions excepted. Values given are industry standards.

Conductivities

Z-Trode

85% IACS
@ 20°C



CuCrZr

80% IACS
@ 20°C

Melting Points

Zinc	~420°C
Brass (70Cu/30Zn)	~1027°C
CuCrZr	~1075°C
Z-Trode	~1080°C
Steel	~1427°C

Electrical and physical constants

Spatter matters

Z-Trode is the electrode of choice for the Spatter Reduction Project. The reduction or even elimination of weld expulsions reduces maintenance, protective clothing and rework costs.

Traceability

All materials are fully traceable. Z-Trode electrodes can be recognized by their flats around the periphery of the electrode.

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